

Ravi Chandrasekar

Senior Research Scientist – Natural Language Processing

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CORE TECHNICAL SKILLS

Machine Learning & NLP: Transformer architectures, fine-tuning, prompt engineering, sequence labeling, information extraction, semantic similarity, evaluation metrics (BLEU, ROUGE, F1, MRR) | **Frameworks & Tools:** PyTorch, HuggingFace Transformers, spaCy, NLTK, JAX, TensorFlow | **Data & Evaluation:** Dataset curation, annotation protocols, benchmark design, statistical significance testing, error analysis | **Infrastructure:** SLURM clusters, GPU scheduling, experiment tracking (Weights & Biases), version control | **Programming:** Python, Bash, SQL, LaTeX

PROFESSIONAL EXPERIENCE

Senior Research Scientist, National Institute for Language Technology Standards

Washington, DC | January 2024 – Present

- Lead development of named entity recognition models for government policy document analysis; achieved 91.2% F1 on internal NILT-Policy-2025 benchmark with 847 manually annotated documents
- Designed and implemented few-shot learning pipeline for transfer across 12 government agencies' distinct entity taxonomies; evaluated on held-out test sets with macro-averaged metrics
- Published "Multilingual NER in Low-Resource Government Domains" in *Proceedings of ACL 2025* (34 citations to date); paper demonstrates 8.3% improvement over prior SOTA on cross-domain adaptation task
- Conducted comparative analysis of 18 open-source LLM architectures for document summarization; published technical report adopted by 6 federal agencies for procurement decisions
- Mentored 2 junior researchers on experiment design and reproducibility standards; established laboratory practices for code archiving and hyperparameter documentation

Postdoctoral Researcher, Center for Advanced Computational Linguistics, Boston University

Boston, MA | August 2022 – December 2023

- Investigated neural architectures for semantic role labeling on PropBank and FrameNet datasets; developed attention visualization toolkit to interpret model decisions
- Co-authored "Probing Tasks Reveal Shallow Representations in Modern Language Models" in *Findings of EMNLP 2023* (127 citations); work challenged assumptions about syntactic knowledge in encoder models
- Established evaluation framework for measuring linguistic knowledge in pretrained models across 14 diagnostic tasks; results published in collaborative benchmark paper with 6 institutions
- Implemented contrastive learning approach for improving out-of-distribution generalization; evaluated on 9 standard datasets with 3.2-6.8% performance gains

PhD Candidate, Department of Computer Science, Carnegie Mellon University

Pittsburgh, PA | September 2018 – July 2022

- Dissertation: "Efficient Architectures for Structured Prediction in NLP" — defended July 2022; examined graph-based decoding methods and their computational trade-offs
- First-author publication "Graph Transformers for Semantic Parsing" in *Proceedings of NAACL 2022* (89 citations); introduced novel hybrid architecture combining graph neural networks with transformer attention

- Developed benchmark suite for semantic parsing across 5 datasets (Abstract Meaning Representation, Semantic Role Labeling, Universal Dependencies); published reproducible evaluation code and detailed error analysis
- Second-author on "Multilingual Semantic Parsing with Shared Latent Structures" in *ACL 2021* (156 citations); contributed architecture improvements and cross-lingual evaluation methodology
- Awarded NSF Graduate Research Fellowship (2019–2022) for research on low-resource parsing
- Presented research at 8 international conferences; peer-reviewed submissions for ICLR 2022, ACL 2021–2023

Research Intern, Allen Institute for AI

Seattle, WA | June 2021 – August 2021

- Evaluated existing question-answering models on SQuAD, Natural Questions, and proprietary academic QA dataset; identified failure modes through systematic error categorization
- Prototyped retrieval-augmented approach for improving answer span prediction; tested on 3 benchmark datasets with ablation studies

Undergraduate Research Assistant, MIT Computer Science and Artificial Intelligence Laboratory

Cambridge, MA | January 2018 – May 2018

- Analyzed linguistic biases in word embeddings (GloVe, Word2Vec); contributed to paper examining gender representation across 8 languages

EDUCATION

PhD in Computer Science, Carnegie Mellon University, Pittsburgh, PA | 2022

- Dissertation advisor: Professor Linda Zhang | GPA: 3.9/4.0

Bachelor of Science in Computer Science & Mathematics, University of Massachusetts, Amherst, MA | 2018

- Summa Cum Laude | GPA: 3.95/4.0

SELECTED PUBLICATIONS & METRICS

- Chandrasekar, R., et al. "Graph Transformers for Semantic Parsing." *NAACL 2022*. [89 citations]
- Chandrasekar, R. "Probing Tasks Reveal Shallow Representations in Modern Language Models." *EMNLP 2023 Findings*. [127 citations]
- Chandrasekar, R., Zhang, L. "Efficient Architectures for Structured Prediction in NLP." *PhD Dissertation, CMU*, 2022.

AWARDS & RECOGNITION

- NSF Graduate Research Fellowship, 2019–2022
- Best Poster Award, NLP Reading Group Symposium, CMU, 2021